
MA 709: Numerical Linear Algebra

Course Code	MA 709
Course Name	Numerical Linear Algebra
L-T-P-C	2-0-2-3
Prerequisites	MA-607 or equivalent, MA-609 or equivalent and IC-111 or equivalent.
Intended for	M.S./Ph.D.
Distribution	Elective
Approval	5th Senate (Oct 2013)

Course Contents

- **Fundamentals**
Overview of matrix computations, norms of vectors and matrices, stability and ill-conditioning, Condition number of a matrix and its applications. [6 Lectures including Lab]
- **Systems of Linear Equations**
Gaussian eliminations with and without Pivoting, Gauss-Seidel, Successive-over-relaxation methods, LU factorization, Cholesky factorization, stability and sensitivity analysis. [9 Lectures including Lab]
- **Eigen value Problem**
Properties of the eigen decomposition, Power's Method, The LR and QR algorithms, Rayleigh quotient iteration, inverse iteration. [10 Lectures including Lab]
- **Eigen values of Symmetric Matrices**
Orthogonal Matrices, Jacobi Method, Givens Method, Gram-Schmidt Process, Householder Method. [9 Lectures including Lab]
- **Singular Value Decomposition**
Properties of the singular value decomposition, Methods for the singular value decomposition. [8 Lectures including Lab]

Laboratory Modules

Each week the lab class will follow the corresponding lecture classes of that week. [2 hours]

Text Books

1. David S. Watkins, **Fundamentals of Matrix Computations**, 2nd Edition, John Wiley & Sons, 2002.
2. L. N. Trefethen and David Bau, **Numerical Linear Algebra**, SIAM, 1997.
3. C. T. Kelly, **Iterative Methods for Linear and Nonlinear Equations**, SIAM, Philadelphia, 1995.

References

1. G. H. Golub and C.F. Van Loan, **Matrix Computation**, 3rd Edition, Hindustan book agency, 2007.
2. B. N. Datta, **Numerical Linear Algebra and Applications**, 2nd Edition, SIAM, 2010.
3. O. Axelsson, **Iterative Solution Methods**, Cambridge University Press, 1994.
4. D. J. Higham and N.J. Higham, **Matlab Guide**, SIAM, (recommended as a Matlab reference).